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ORIGINAL PAPER

Enhancing autumn greenway walking experience: Exploring the combined effects of noise and thermal environment

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Abstract

The evaluation of outdoor green spaces is influenced by diverse sensory perceptions. Traffic noise and thermal conditions significantly impact greenway-walking satisfaction; their optimization is vital for improving user experience and encouraging outdoor engagement. The study examines a typical Beijing greenway during autumn, focusing on strategies to enhance the walking experience under the combined effects of noise and thermal environments through mobile measurements and surveys. The results show that: 1) The interplay between noise and thermal factors varies depending on the walking state. Upon arrival, an increase in noise significantly worsens thermal comfort; higher sound levels intensify warm thermal sensations, though this effect is not consciously perceived. Upon departure, the effect of noise on thermal perception is not obvious. In both walking states, thermal sensation significantly affects subjective noise perception, yet the trends of influence differ. Subjective noise loudness increases as thermal comfort worsens, showing significant correlation only upon departure. 2) During autumn greenway walks, acoustic factors exert a greater impact on Overall Environmental Satisfaction (OES), with subjective noise loudness being more influential than noise level, followed by air temperature (T_a). Greater noise decreases OES, while OES increases initially with T_a and then decreases. The integrated effects of noise-thermal factors on OES show significant changes. 3) To enhance the autumn greenway-walking experience, the advised parameters are A-weighted Sound Level (ASL) $\leq$ 59.12 dBA and 15.17 °C $\leq T_a \leq$ 18.75 °C. Finally, three design strategies are proposed: reducing subjective noise loudness, differentiating design based on walking status and balancing acoustic-thermal perceptual preferences.

Keywords Noise · Thermal perception · Greenway walking · Overall environmental satisfaction · Design strategies

Introduction

Urbanization exacerbates a range of environmental issues, such as noise pollution and urban heat island effects (Mohammadzadeh et al. 2023). Noise pollution is one of the foremost environmental challenges faced by humankind (UN Environment Programme, 2022). Traffic noise is a primary source of environmental noise (Khomenko 2022), and a major cause of discomfort in urban green spaces (Liu et al. 2018a; Baquero Larriva and Higueras 2020; Li et al. 2022; Nitidara et al. 2022). Prolonged exposure to high noise levels can impair human health, leading to hearing impairment, ischemic heart disease (Passchier-Vermeer and Passchier 2000) and depression (Eze et al. 2020). Concurrently, adverse thermal conditions can increase the incidence of obesity and kidney diseases (Yu and de Dear 2022; Torres and Dixon 2022) and decrease the frequency of outdoor activities (Mi et al. 2020). Therefore, improving the outdoor acoustic and thermal environment is crucial for promoting physical activity and enhancing overall well-being (Baquero Larriva and Higueras 2020).

Greenways serve as one of the primary venues for residents to engage in physical activities (Yang et al. 2024), known for their linearity and interconnectivity (Liu et al. 2018b). Behaviorally, greenways are primarily ideal for and utilized in supporting slow-moving sports such as walking,

Highlights
- Explored the combined impact of noise and thermal factors to improve greenway walking
- The interplay between noise and thermal factors varies depending on walking state
- Noise subjective loudness, levels, and T_a impact Overall Environmental Satisfaction
- Significant changes in Overall Environmental Satisfaction occur around 55-60 dBA
- Proposed the ranges of noise level and air temperature and three design strategies

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jogging, and cycling (Tan et al. 2019a, b; Zhang et al. 2022). Spatially, while some greenways are nestled within the relative quiet of parks, others are built along urban roads and thus are subjected to higher levels of traffic noise (Chang et al. 2020; Lu et al. 2024), significantly impairing the comfort of the greenways (Bian et al. 2021). Therefore, reducing noise and improving the thermal environment is crucial for enhancing the usage of urban greenways.

The perception of noise and its negative impacts in urban green spaces has been well-discussed. Research shows that participants who perceive lower levels of background noise are more likely to report enhanced place attachment and well-being (Chang et al. 2020). Within natural greenway areas, traffic noise is found to be more irritating than other artificial sounds (Bian et al. 2021). A research indicates that green spaces with elevated noise levels exhibit a notably reduced frequency of walking activity (Rey Gozalo et al. 2018). However, existing research on greenways remains limited, with a dearth of studies examining the dynamic perception of noise and its impact throughout the entire journey of greenway pedestrians.

Studies have investigated the impact of microclimate parameters on thermal comfort in greenways in their local climates. For instance, in Shenzhen's hot and humid climate, wind speed and solar radiation were crucial to pedestrians' thermal comfort on the coastal mountainside greenway (Liu et al. 2022). In northern China, a correlation was observed between perceived wind, humidity, and thermal perception (Cong et al. 2022). In Guangzhou, thermal perception was influenced by shade levels, with pedestrians showing a greater need for shade than cyclists (Jiang et al. 2021). Nevertheless, research on thermal perception in greenways is less abundant than for other green spaces, and often relies on static observations or random surveys This approach often overlooks the dynamic and continuous nature of activities such as walking within greenways. Additionally, studies predominantly focus on summer heat discomfort, neglecting thermal perception during the transitional seasons of spring and autumn. Since people tend to increase their outdoor activities in public spaces during these seasons, ensuring comfort is essential for prolonging outdoor engagement and improving health (Zhang et al. 2024).

A wealth of studies has demonstrated that acoustic and thermal perceptions interact (Pellerin and Candas 2003; Nagano and Horikoshi 2005; Yin et al. 2022; Geng et al. 2022; Nitidara et al. 2022; Mohammadzadeh et al. 2023). Noise has been shown to reduce outdoor thermal comfort (Jin et al. 2020a; Lau and Choi 2021; Yin et al. 2022; Geng et al. 2022), and its impact on thermal sensation varies with the urban space, type of noise, and season (Geng et al. 2022). Concurrently, the thermal environment's influence on subjective acoustic evaluations is linked to meteorological parameters (Yu and Kang 2009). For example, the Harbin square study reveals that acoustic discomfort related to traffic noise is intensified by extreme winter cold and summer heat (Jin et al. 2020a). Despite this, existing research has predominantly concentrated on urban environments such as parks (Baquero Larriva and Higueras 2020; Ren 2023), squares (Jin et al. 2020a; Ren 2023), university campuses (Yin et al. 2022; Geng et al. 2022), and residential areas (Gao 2013), with comparatively less attention given to greenways.

Considering the varied construction methods of greenways, individuals are frequently exposed to complex and shifting acoustic and thermal conditions along these routes. These dynamic conditions, particularly during the act of walking, lead to more intricate perceptions compared to stationary activities in other green spaces (Xie et al. 2022). Furthermore, pedestrians often exhibit "stop-and-go" behavior during strolls (Vasilikou and Nikolopoulou 2013). Thermal perception varies between the states of walking and being stopped, with meteorological parameters influencing these perceptions differently (Peng et al. 2022). There is a gap in research regarding the potential differences in interaction effects between subjective perceptions of noise and thermal environments in real greenway environments across different walking states.

Individuals' overall comfort in outdoor spaces is influenced by both direct acoustic and thermal impacts and their interactions, particularly among sound types, levels, and temperature (Jin et al. 2020a, b; Yin et al. 2022; Chen et al. 2022a). Moreover, in various green spaces such as residential areas, squares, and university campuses, the influence of acoustic and thermal environments on overall comfort differs (Gao 2013; Jin et al. 2020a; Yin et al. 2022). Currently, research into the combined effects of noise and thermal factors on overall environmental satisfaction within greenways is scarce. The independent and combined impacts of noise and thermal parameters on satisfaction in greenway environments require more in-depth investigation.

While some studies on greenway acoustic and thermal environments have proposed corresponding design strategies, they tend to focus on a single perceptual improvement. From a thermal perspective, the use of WP bricks and tiles is advocated in the design of Guangzhou's greenways, with a particular emphasis on the advantages of arbors and water elements for shading and promoting evaporative cooling (Li et al. 2013). Concurrently, research in Beijing has demonstrated the superior cooling and humidifying effects of arboreal-grassy communities and asphalt within greenway environments (Li 2022). From an acoustic standpoint, a 5-m depth of vegetation has been identified as the optimal barrier for mitigating traffic noise (Ow and Ghosh 2017). Additionally, the method of using devices to play natural sounds to mask noise is also recommended (Van Renterghem et al.

2020). However, there is a lack of combined greenway design strategies to meet the dynamic acoustic and thermal needs of pedestrians in complex real-world greenway environments.

Therefore, improving the acoustic and thermal environments to enhance greenway walking experience is essential. This study explores the interactive effects of noise and thermal factors on greenways based on real autumn walking experiences, their impact on overall environmental satisfaction, and develops practical greenway design strategies for urban designers. Consequently, the study's objectives are delineated as follows:

1) Investigating the impact of noise and thermal objective parameters and their subjective perception on each other's subjective perceptions during different walking status on the greenway in autumn.
2) Explore the impact of objective noise and thermal parameters and their subjective perceptions on overall environmental satisfaction and the combined effect of both on it during the whole process of autumn greenway walking.
3) Considering complex greenway environments, delineate the regulatory ranges for objective noise and thermal parameters in autumn and propose practical planning and design strategies to enhance the walking experience on greenways.

Methods

The site

This study was conducted in Beijing, a typical city with a warm temperate continental monsoon climate in China (Yan et al. 2022). From 1981 to 2010, Beijing's average July temperature was recorded at 27.3 ℃, while January averaged -3 ℃ (https://data.cma.cn/) Autumn is typically a season of pleasant weather in Beijing, with moderate temperatures, making it the most frequent season for outdoor activities among Beijing residents.

The Three Mountains and Five Gardens Greenway is one of the typical greenways in Beijing (Tan et al. 2019b), and it is also a main greenway chosen by residents for daily leisure activities. Before formal experimentation, a survey of typical greenways in Beijing identified notable variations in noise and thermal environments across different greenway spaces. Park-located sections typically exhibit lower noise levels, with their thermal profiles largely shaped by surrounding vegetation. In contrast, sections bordering urban roads experience greater traffic noise influence and have thermal environments affected by multiple factors. This study examined representative greenway sections, typical of those used by residents for autumn outings, characterized by the aforementioned environmental features and totaling approximately 5.55 kilometers in length. P1-P5, proximate to or located on municipal roads, experience higher noise impact; P7-P11, situated within parks, are less affected. Point P6, though on a municipal road, is near a park and is subject to combined environmental influences. The environmental characteristics and the distribution of stopping points are shown in Figure 1 (a-c).

Field experiment

Experimental procedure

A total of 22 respondents (11 of each gender) joined this study. The characteristics summary of these 22 participants can be found in SI-Table 1 in supplementary material named SI-TABLE. The greenway walking experiment was conducted on October 25 and 31, 2022, in Beijing, capturing typical autumn weather with clear skies and minimal cloud cover, while avoiding windy or rainy conditions. The experiment focused on the peak usage times of 10:00-13:00 and 13:30-16:30, ensuring that the experiment captured representative data of the greenway's usage patterns in ideal weather conditions.

Figure 1 (d) illustrates the procedure of this experiment. The focus of this study is on the most common walking activity in greenways, employing a "stop-and-go" approach to investigate whether there are differences in the interactive effects of noise and thermal factors when individuals first arrive at a space and when they are about to depart after a rest. The method replicates residents' greenway touring patterns, with measurement points spaced according to typical walking distances from preliminary investigation. Referring to Peng Z et al.'s climate walking experiment (Peng et al. 2022), the resting time at each stop was set to 5 minutes, which is a time interval sufficient to distinctly differentiate the impacts produced by these varying walking states. The questionnaire method was utilized to explore the subjective evaluations, and the physical environment was measured while completing the questionnaire.

During the warm-up phase, the respondents rested in P1 for 30 min to adapt to the outdoor environment. In the walking phase, respondents started walking at a speed of 1.4 m/s from P1. After arriving at each stopping point, they stayed in place for 5 minutes before starting again until reaching P11. Respondents were required to fill out the questionnaire when they had just reached the stopping point and stayed in place for 5 minutes. The instruments were moved alternately during the walking phase to ensure that it was in a steady state when respondents reached each stopping point. The walking distance between each point was similar to avoid different

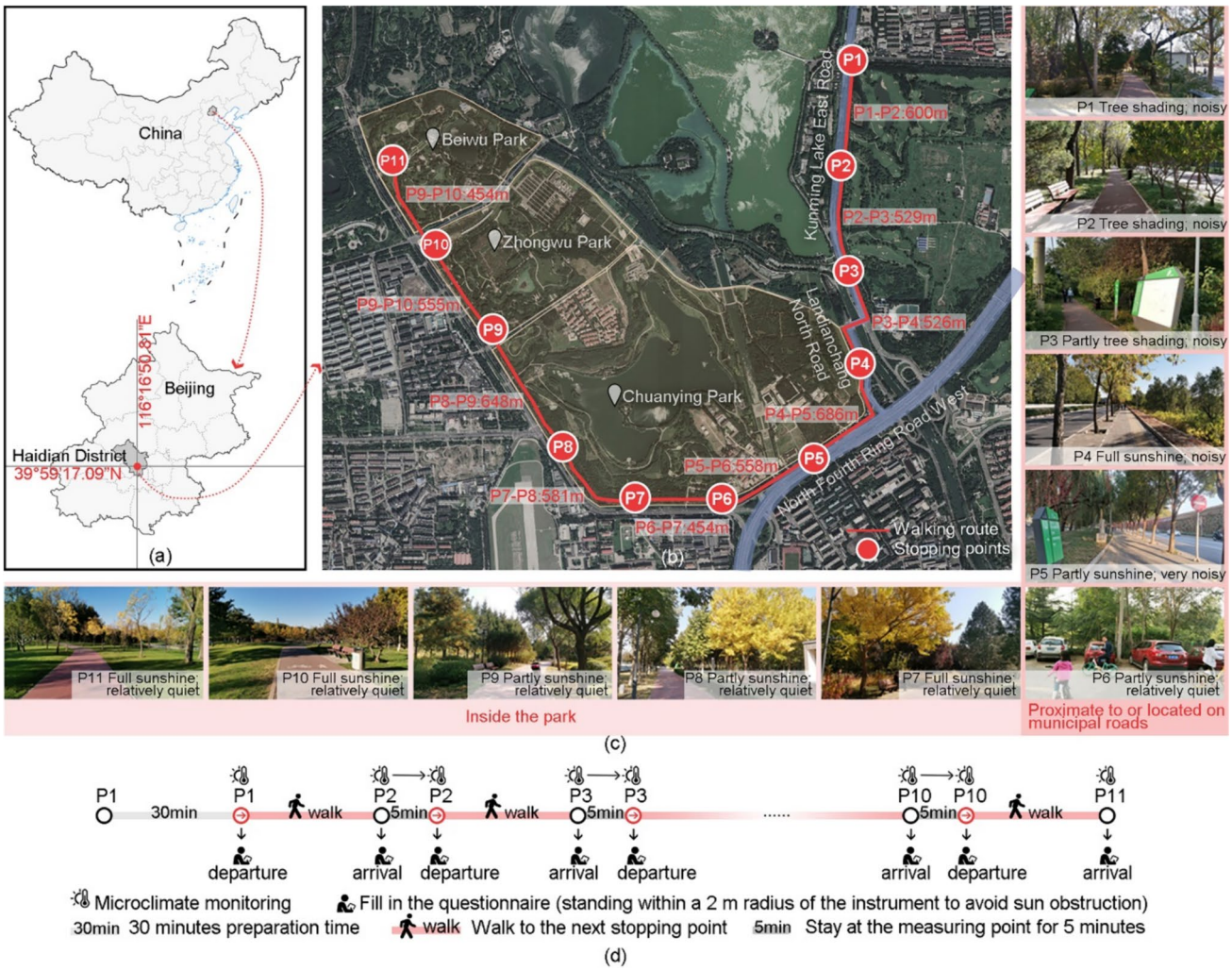


Fig. 1 **a** Study area; **b** Greenway walking route and the distribution of stopping points; **c** Environmental characteristics of the stopping points; **d** Experimental procedure

states of the respondents at each measurement point due to uneven walking distances.

Measurement of physical parameters

The technical details of the monitoring system are listed in Table 1. It collected air temperature (T_a), relative humidity (RH), and wind speed (V_a) globe temperature (T_g) and ASL at 1.5 m above the ground at each stopping point. The Mean Radiant Temperature (MRT) is calculated by Eq. (1). Selection of measured instruments and calculation of MRT refer to ISO 7726 standard (ISO 1998). The microclimate measuring instruments were calibrated before the start of the formal experiment.

$$MRT = \left[\left(T_g + 273.15\right)^4 + \frac{1.1 \times 10^8 \times {v_a}^{0.6}}{\varepsilon_g \times {diam_g}^{0.4}} \times \left(T_g - T_a\right) \right]^{0.25} - 273.15 \quad (1)$$

Table 1 Measured parameters and instruments

Equipment	Parameter	Range	Accuracy	Sampling rate
HOBO onesite U23-001	T_a	-40–70 ℃	±0.21 ℃	1 min
	RH	0–100%	±2.5%	
Delta OHM HD32.3TC	T_g	-30–120℃	1/3 DIN	
	V_a	0.05–5 m/s (0–80℃)	±(0.05 + 0.5% measure) m/s	
SMART SENSOR AS-824	ASL	30 ~ 130dBA	± 1.5 dBA	manual recording

where $diam_g$ is globe diameter ($diam_g = 0.05$ m in this study) and ε_g is emissivity ($\varepsilon_g = 0.95$ for a black globe).

Questionnaire survey

This study explored respondents' subjective evaluations of the acoustic and thermal environments, as well as their overall satisfaction of the environment. The Thermal Sensation Vote (TSV) was assessed using the ASHRAE 9-point scale, while the Thermal Comfort Vote (TCV) was examined with a conventional 4-point scale, chosen for its enhanced granularity in capturing the nuances of thermal discomfort. These scales have been employed in prior research (Li et al. 2016). Since Subjective Loudness Vote (SLV) is essential for assessing the acoustic environment (Ren 2023), it was evaluated in this study using a 5-point scale. The Overall Environmental Satisfaction (OES) is calculated based on the Landscape Aesthetics Vote (LAV), Landscape Enjoyment Vote (LEV) and the vote on the extent to which current greenways are chosen as play destinations (ECV), as these factors are closely related to visitors' overall evaluation of the environment and can influence their greenway experience and frequency of visits (Niu et al. 2022; Yan et al. 2023). LAV, LEV, and ECV were collected by using a 5-point scale. In total, 438 effective survey samples were selected.

Statistical analysis

Entropy method is an objective weighting method that determines the weights of indicators according to the information entropy and judges the degree of influence of the indicator on the comprehensive evaluation. In this study, the evaluation of OES is generated by calculating the weights of LAV, LEV, and ECV. To eliminate the influence of the scale on the calculation results, the data were normalised before analysis.

Column charts were used to descriptive the subjective thermal perception and perceived acoustic quality. Linear and quadratic functions were used to investigate the relationship between thermal variables (TSV/TCV/meteorological parameters) and acoustic variables (ASL, SLV), and their respective effects on OES. Nearest neighbor interpolation and thin plate spline smoothing method were employed to generate a matrix, providing a visual contour map that demonstrates the comprehensive impact of acoustic and thermal parameters on OES.

Results

Microclimate and noise conditions

Meteorological parameters and sound level measured in 11 stopping points are listed in Table 2. Average T_a ranged from 16.56 ℃ to 18.58 ℃, maximum and minimum T_a was recorded respectively at P5 (19.36 ℃) and P1 (14.00 ℃). Overall, P1-6 showed a greater temperature difference. Average RH ranged from 34.15% to 38.99%, demonstrating a relatively stable humidity condition. Wind speed (V_a) was low in most cases with the average V_a ranging from 0.6 m/s (P9) to 1.35 m/s (P5), while V_a could reach up to 3.51 m/s at P10. Average MRT ranged from 20.57 ℃ (P1) to 49.71 ℃ (P3), while maximum and minimum value reach up to 65.30 ℃ (P7) and 14.58 ℃ (P11), demonstrating the variable radiation environment along the route.

The average sound levels at P1-P4 surpass 60 dBA, with P5 reaching a peak of 75.75 dBA, whereas most other stopping points maintain average sound levels no higher than 55 dBA. This indicates that greenways in proximity to municipal roads are subject to higher noise pollution, in contrast to those located within park settings, which tend to be more tranquil.

Acoustic and thermal subjective response

Distribution of TSV, TCV, SLV at each stopping point

The distribution of TSV and TCV during the greenway walk is shown in Fig. 2 (a-b). The combined proportion of "Slightly warm" and above (TSV > 0) for P3, P4 was quite high, at 88.6%, 86.4%, respectively, followed by P6 (64.4%), P5 (61.9%), and P7 (58.1%). While the sum of proportions of "Slightly cold" and below in P1 was 71.4%. The vote of "Uncomfortable" only reported at P1 (4.8%), P5 (4.8%), P2 (4.4%) and P3 (2.3%), with P3 showing the lowest proportion of "Comfortable" at 54.6%. Throughout

Table 2 Meteorological parameters and sound level along the experiment route

Stopping points	Max. T_a (℃)	Avg. T_a (℃)	Min. T_a (℃)	Max. RH (%)	Avg. RH (%)	Min. RH (%)	Max. Va (%)	Avg. Va (%)	Min. Va (%)	Max. MRT (℃)	Avg. MRT (℃)	Min. MRT (℃)	Max. SL (dBA)	Avg. SL (dBA)	Min. SL (dBA)
P1	17.84	16.56	14.00	48.84	38.99	19.30	1.35	1.08	0.94	23.35	20.57	15.02	63.00	62.60	62.40
P2	19.29	17.10	14.36	47.00	36.84	18.55	1.45	0.73	0.26	51.69	24.19	17.45	69.70	64.37	59.00
P3	19.29	18.20	15.73	44.86	34.31	17.53	1.75	1.11	0.44	62.99	49.71	24.26	72.50	64.79	56.30
P4	19.30	17.97	15.48	43.79	34.15	16.70	2.16	1.31	1.17	58.04	36.72	24.26	73.70	61.18	56.30
P5	19.36	18.42	16.67	43.49	34.41	16.41	2.71	1.35	0.55	60.49	35.52	22.86	77.00	75.75	73.10
P6	19.15	17.63	15.31	47.45	37.78	20.03	1.12	0.65	0.18	41.87	25.87	20.07	56.20	47.97	37.80
P7	19.17	18.58	17.36	44.44	34.40	17.13	1.45	0.99	0.68	65.30	41.31	29.90	56.20	55.07	54.00
P8	18.93	18.05	16.51	46.84	34.88	16.01	2.18	1.17	0.23	58.04	28.53	18.78	73.70	52.82	47.40
P9	18.67	17.75	16.30	46.84	34.89	16.30	1.22	0.60	0.23	31.84	22.82	18.78	54.20	50.00	47.40
P10	18.52	18.20	17.57	46.44	36.56	16.17	3.51	1.15	0.51	61.14	30.45	17.19	47.50	43.98	42.50
P11	17.77	17.63	17.38	49.20	37.04	18.01	1.29	0.86	0.62	52.40	29.36	14.58	43.80	40.69	39.20

the entire walking process, no respondents felt "Very uncomfortable".

Figure 2 (c) illustrates the distribution of SLV among stopping points. The sum of the proportions of "Medium (3)" and above for P1-P4 ranged from 71.43% to 95.45%. At P5, the sum of proportions of "More often (4)" and "Extremely much (5)" reached up to 92.86%. Respondents perceived less noise at P6-P11, with the sum of the proportions of "Hardly (1)" and "Relatively few (2)" ranging from 41.67% to 78.26%.

Distribution of SLV under TSV, TCV classification

Figure 2 (d-e) illustrates the distribution of SLV among TSV and TCV. No respondents reported "Extremely much (5)" when they felt "Cold" and "Cool". The percentage of extreme votes for subjective noise loudness is higher when the thermal sensation is hotter. When the thermal sensation was reported as "very hot", the proportion of "Hardly (1)" and "Extremely much (5)" was noticeably higher.

Respondents who labeled their thermal comfort as "Uncomfortable" were more prone to assess the noise level as "More often (4)" or "Extremely much (5)", accounting for 66.6% of the votes. This figure is markedly higher than the aggregate percentage of the same noise ratings within the "Slightly uncomfortable" (34.6%) and "Comfortable" (27.7%) thermal comfort categories.

Distribution of OES at each stopping point

The weight results calculated using the entropy method. ECV (35.38%) has a slightly higher weight in the influence of OES than LEV (32.89%) and LAV (31.73%) and is the most influential factor. Detailed information on the determination of weights can be found in SI-Table 2 in supplementary material named SI-TABLE.

Figure 2 (f) illustrates the average OES for each stopping point. P11 exhibited the highest mOES at 0.94, peaking just below the maximum of 1, whereas P5 recorded the lowest with a score of 0.13. P6-P10 maintained average satisfaction scores above 0.5, the median threshold, while P1-P4 fell below 0.4. This distribution of scores parallels the distributional characteristics of SLV in Fig. 2 (c).

Interaction between acoustic and thermal factors

Effects of ASL and SLV on TSV, TCV

Using the Bin method, Fig. 3 (a-d) depicts the relationship between mean SLV or ASL at every 1-unit or 5-dBA

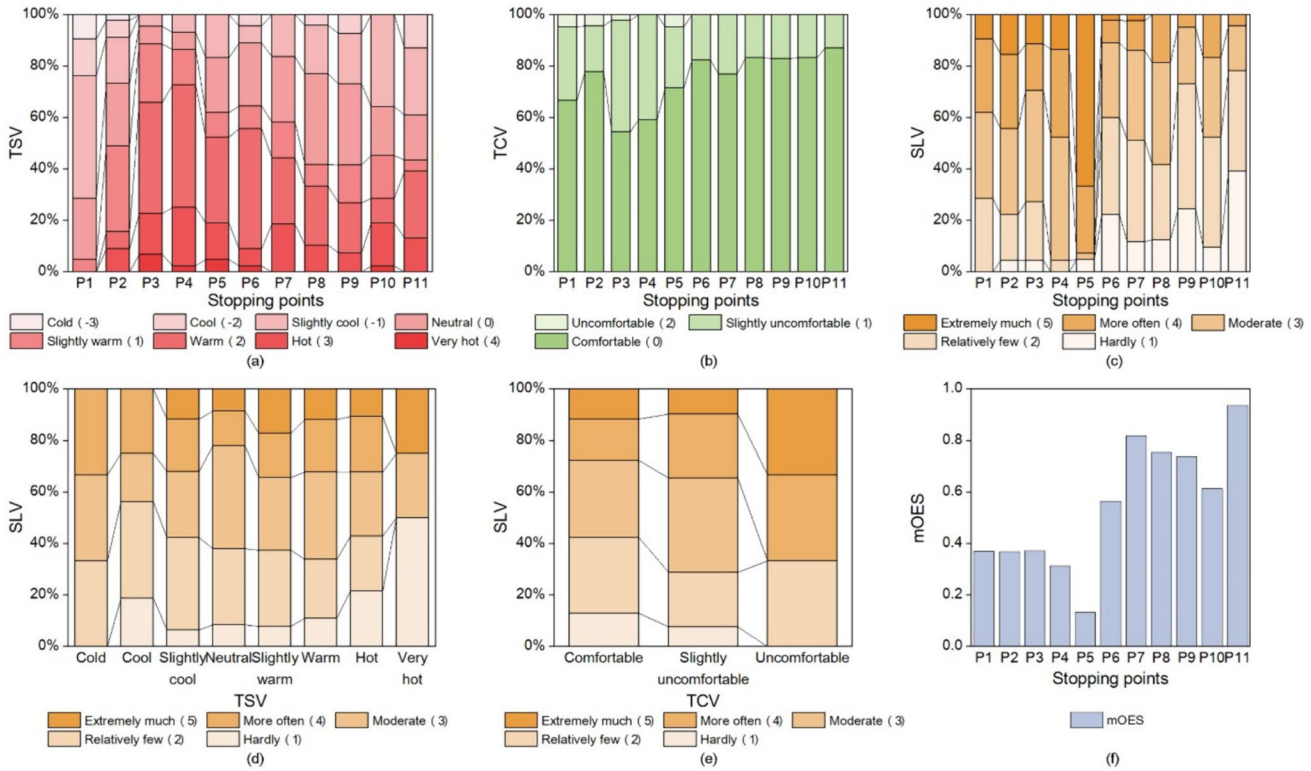


Fig. 2 Distribution of the TSV (**a**), TCV (**b**) and SLV (**c**) among stopping points; Distribution of SLV among TSV (**d**) and TCV (**e**); (**f**) Average OES for each stopping point

interval and the corresponding TSV/ TCV. The results indicate that the impact of ASL on TSV/ TCV and SLV on TCV in the arrival state is statistically significant ($p < 0.01$, $p < 0.05$ or $p < 0.1$). Heightened sound levels are associated with a heightened thermal sensation and diminished thermal comfort. While an increase in subjective noise loudness exacerbates thermal discomfort, it does not significantly impact thermal sensation. In the departure state, increases in ASL and SLV tend to make the thermal sensation cooler and thermal comfort worsen, but these effects are not statistically significant.

Effects of TSV, TCV and Meteorological parameters on SLV

Figure 3 (e–f) describes the relationship between mean TSV or TCV at every 1-unit interval and the corresponding SLV. The curve of TSV fitted to SLV was significant at the 0.05 level. In the departure state, when $TSV \geq -0.47$, subjective noise loudness decreases as thermal sensation increases. Conversely, in the arrival state, when $TSV \leq 2.38$, subjective noise loudness gradually increases as thermal sensation escalates. SLV tends to become larger as thermal comfort gets worse in both states, but this tendency is statistically significant (at the 0.1 level) only in the departure state.

SI-Table 3 (see supplementary material named SI-TABLE) describes the relationship between mean T_a, RH, V_a or MRT at every 1 ℃, 1%, 0.1 m/s or 1 ℃ interval and the corresponding SLV. The results show that the fit of the four meteorological parameters to the SLV is not statistically significant in both states, indicating that the meteorological parameters do not affect the subjective loudness of the noise.

Effects of acoustic and thermal factors on OES

Figure 4 (a-d) describes the effect of TSV (a), TCV (b), SLV (c), ASL (d) and T_a (e) on OES. The fitting results for SLV and ASL on OES were statistically significant at the 0.01 level, whereas the fit of TSV and TCV on OES were not significant. In addition, the statistical fit of ASL ($R^2 = 0.5304$) is less than of SLV ($R^2 = 0.7228$), indicating that SLV had more explanatory power for OES. Among the meteorological parameters, only the fit of T_a to OES was significant ($p < 0.05$). The relationships between other microclimate parameters and OES are presented in SI-Table 4 (see Supplementary Material SI-TABLE).

Figure 4 (f-k) demonstrates the combined effect of subjective and objective acoustic-thermal parameters on OES. From a subjective standpoint, higher OES can be achieved when SLV is around 2–2.5 (2-Relatively few, 3-Moderate),

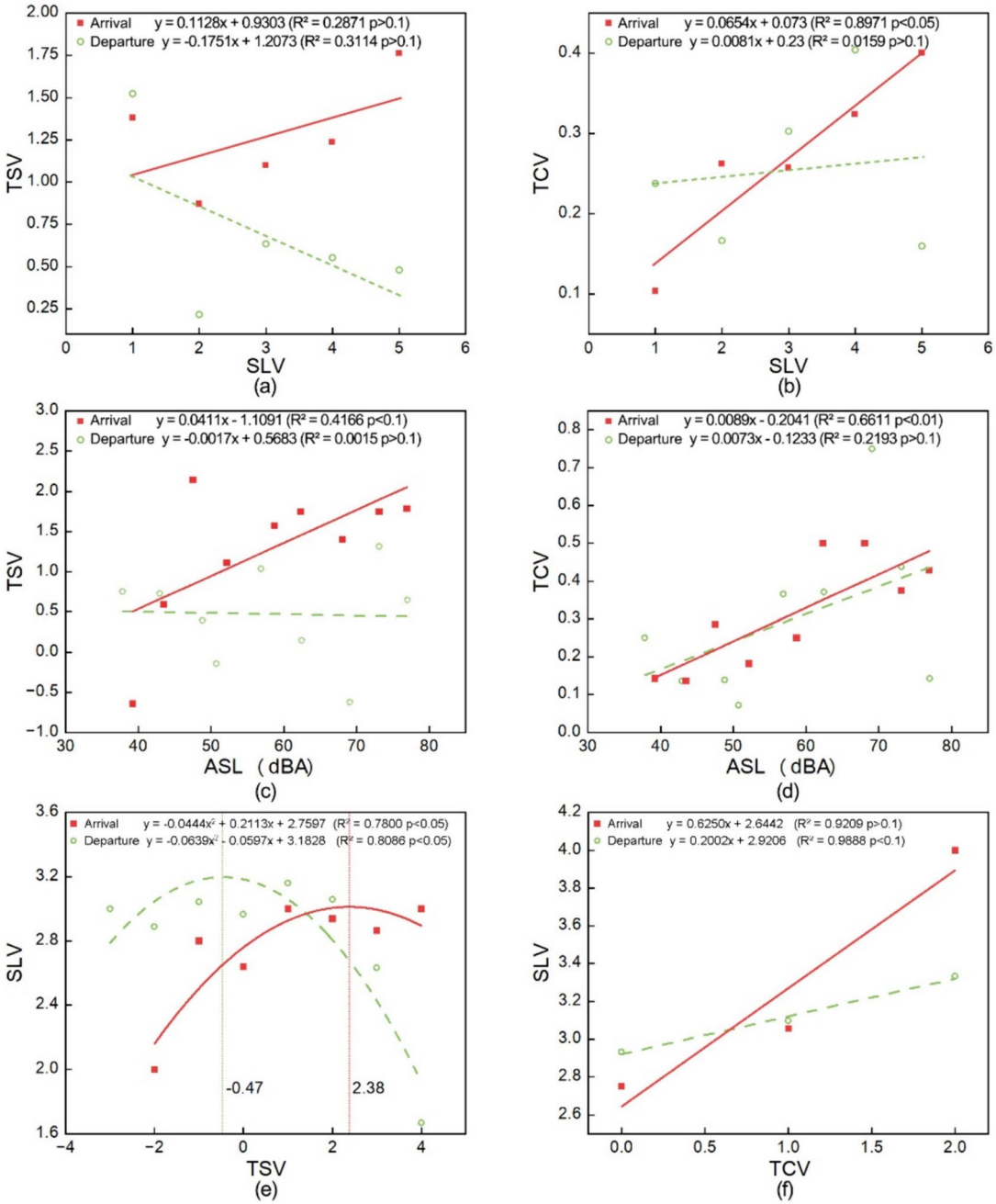


Fig. 3 Models of the effect of SLV on TSV (**a**), TCV (**b**); models of the effect of ASL on TSV (**c**), TCV (**d**); Models of the effect of TSV (**e**), TCV (**f**) on SLV

TSV is around 2 (warm) and TCV is low (close to thermal comfort). Objectively, the isopleths of OES exhibit greater density around the value of 0.5, notably within the 55–60 dBA noise level range. This indicates that significant changes in overall environmental satisfaction occur near this boundary. Consequently, the calculated ASL and T_a ranges for achieving an OES of 0.5 or above (OES range 0–1) are ASL ≤ 59.12 dBA and 15.17 ℃ ≤ T_a ≤ 18.75 ℃, as shown in Fig. 4 (d-e).

Discussion

Environmental evaluations under noise and thermal influences

Effects of noise on thermal perception

The influence of noise on thermal sensation has shown variability in previous studies. In Harbin's urban squares,

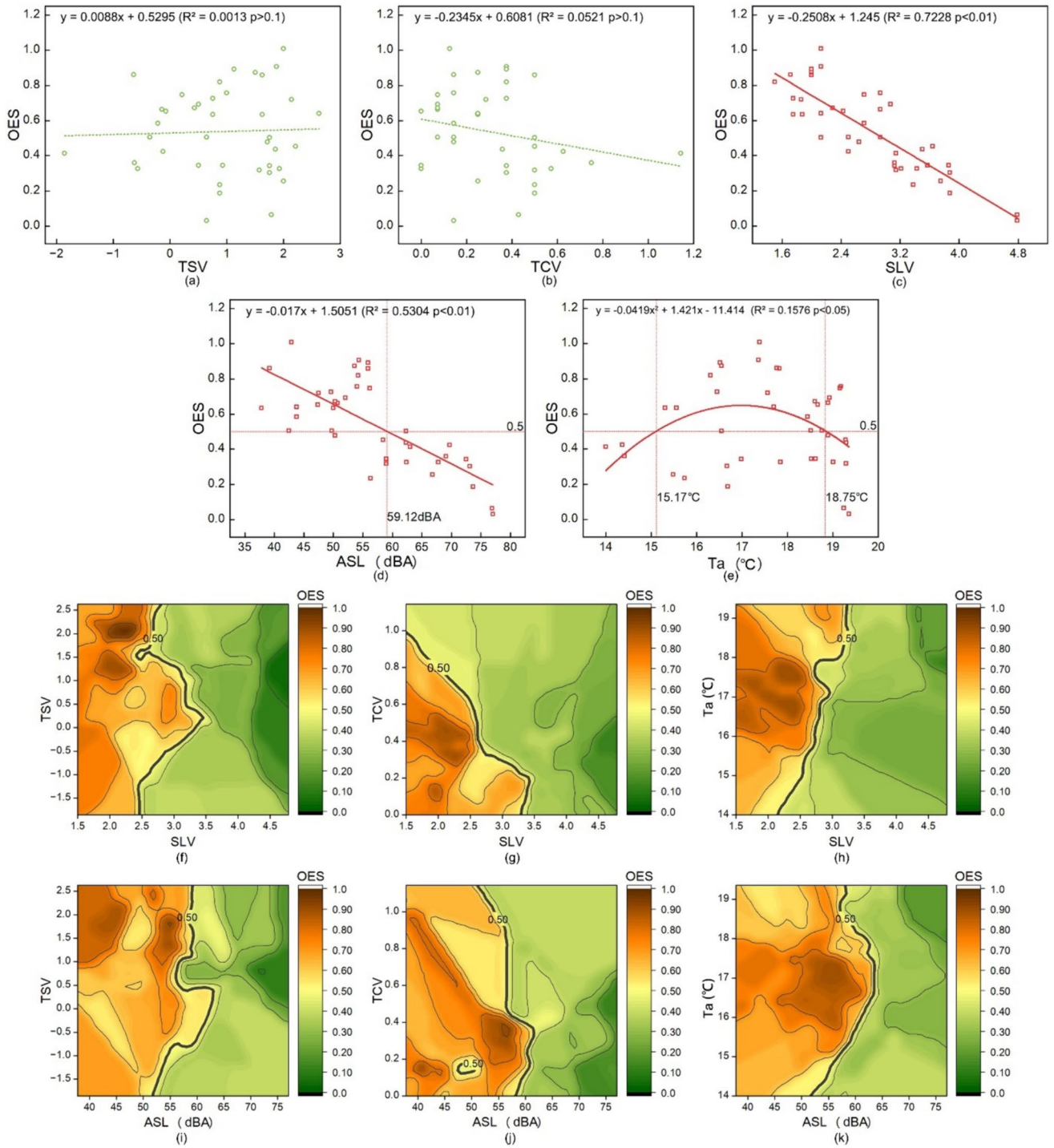


Fig. 4 Models of the effect of TSV (**a**), TCV (**b**), SLV (**c**), ASL (**d**) on OES; (**e**) Models of the effect of T_a on OES; (**f**-**k**) The combined effect of subjective and objective acoustic-thermal parameters on OES

noise levels do not affect the thermal sensation of users during transitional seasons (Jin et al. 2020a). However, in Xi'an's university campuses, noise levels influence the thermal sensation in autumn (Yin et al. 2022). Specifically, at a low sound pressure level (40–50 dBA), noise can amplify the hot sensations in warmer conditions and cold sensation in cooler settings. This study further explores the impact of noise on thermal sensation under different walking states. In the departure state, an increase in subjective noise loudness tends to make the thermal sensation cooler,

but this trend is not statistically significant. In the arrival state, although the fitting results between subjective noise loudness and thermal sensation votes are not significant, the fitted results in Fig. 3 (c) demonstrate that as noise sound level increases, thermal sensation is correspondingly heightened, and it was statistically significant. This indicates that, despite the significant variation in thermal sensation among people during transitional seasons (see SI-Table 5 in supplementary material SI-TABLE), which aligns with the results from studies on urban squares in Harbin (Jin et al. 2020a), noise still impacts subjective thermal sensation in the arrival state, even if this influence is not consciously perceived. Therefore, greenway design must consider the dynamic nature of the potential impact of noise on thermal perception.

Previous studies have indicated that respondents dissatisfied with the acoustic environment are more likely to perceive increased thermal discomfort (Lau and Choi 2021), and heightened noise levels worsen thermal comfort (Wu 2004; Nagano and Horikoshi 2005; Yang and Moon 2018; Guan 2020; Jin et al. 2020a, b), particularly during transitional seasons (Jin et al. 2020a, b; Yin et al. 2022). Our study reveals that the effect of noise on thermal comfort during greenway walks is related to the walking state, with a statistically significant effect on TCV observed only upon initial arrival. This adverse influence diminishes after a period of stay, suggesting an adaptive adjustment to environmental conditions, corroborating the findings by Baquero Larriva et al. in their examination of outdoor spaces in Madrid (Baquero Larriva and Higueras 2020).

Effects of thermal factors on noise loudness perception

During greenway walking, thermal sensation significantly influences the subjective perception of noise loudness, but this perception varies with different walking states. In the departure state, the subjective loudness of noise quickly decreases with the increase of thermal sensation when TSV ≥ -0.47. However, in the arrival state, the subjective loudness of noise increases as the thermal sensation becomes warmer until it reaches TSV of 2.38, after which it slowly declines (Fig. 3-e). This phenomenon clarifies why Fig. 2 (d) shows a higher incidence of extreme votes in subjective noise loudness when thermal sensation is on the hotter side, as the perception of noise loudness differs among respondents experiencing the same level of heat but in different walking states.

Geng et al.'s study reported that participants perceived noises as louder under conditions of thermal comfort or thermal discomfort (Geng et al. 2022). In contrast, our findings show that subjective noise loudness is minimal in thermal comfort states and increases with the rise in thermal discomfort. While this trend is observed in both the departure and arrival states, it only statistically significant in the departure state.

Additionally, while Yu et al. identified air temperature, wind speed, and relative humidity as key factors affecting noise evaluation in urban spaces (Yu and Kang 2009), our study found no impact of meteorological parameters on noise loudness perception.

Combined noise-thermal effects on overall environmental satisfaction

This research examines the separate influences of noise and thermal factors on overall environmental satisfaction, along with their joint impact. In the analysis of individual parameter impacts, SLV has the strongest explanatory power for OES, followed by ASL and T_a. Elevated traffic noise levels and subjective loudness correlated with reduced overall satisfaction, while the effect of air temperature on overall satisfaction follows an inverted U-shaped quadratic function. Previous studies have similar findings. The overall comfort in Harbin's public squares decreases with increased traffic noise during transitional seasons (Jin et al. 2020a). Similarly, subjective comfort among users in Xi'an's campuses is reduced with higher noise levels, with a more pronounced effect in high and low temperature conditions (Yin et al. 2022).

Rey Gozalo et al.'s research in Cáceres, Spain, revealed that activities such as walking are more frequently affected by noise, and this impact has the highest correlation with overall annoyance levels (Rey Gozalo et al. 2018). This study further found that the combined impact of noise and thermal factors on overall environmental satisfaction in greenway walking experiences significant changes around OES = 0.5, particularly within the 55–60 dBA range (Fig. 4 f-k). This phenomenon may be attributed to variations in greenway construction approaches. Stopping points with sound levels below 55 dBA are predominantly located within parks (Table 2), while those with sound levels above 60 dBA are adjacent to municipal roads. The recommended ranges for noise control and air temperature regulation are ASL ≤ 59.12 dBA and 15.17 ℃ ≤ T_a ≤ 18.75 ℃, respectively. This is because when ASL and T_a are within the ranges that correspond to an OES score of 0.5 or higher, even minor adjustments are highly likely to substantially elevate the OES to a higher level. Additionally, strategies aimed at enhancing the acoustic and thermal environment of greenways to increase walking satisfaction should be adjusted to account for the differences in the locations where greenways are constructed.

The extent to which acoustic and thermal comfort influence overall thermal comfort has varied across studies. Yin et al., in their comparison of university campuses in Xi'an (Yin et al. 2022), residential areas in Shenzhen (Gao 2013), urban squares in Harbin (Jin et al. 2020a), and laboratories

in Qingdao (Zhang 2016), suggested that thermal factors outweigh acoustic factors by 3–4 times in comfort assessments. Conversely, research on parks and squares in Indonesia (Nitidara et al. 2022) indicated that auditory factors have a greater impact on the comfort than thermal factors. In this research, overall, acoustic factors exert a more significant influence on the overall satisfaction during greenway walks and reducing the psychological perception of noise loudness is paramount in enhancing autumn greenway walking experiences.

Greenway design strategies for optimizing acoustic and thermal environments

Human evaluation of the external physical environment is the result of a multisensory perception process, where sensory inputs interact to collectively affect overall comfort (Jin et al. 2020b; Nitidara et al. 2022). In highly urbanized city spaces, it is challenging to independently optimize a single aspect for optimal comfort (Nitidara et al. 2022). Designers can leverage multisensory interactions to compensate for less favorable conditions, achieving a target level of overall comfort (Nitidara et al. 2022), which is vital for the enhancement of greenway environmental design (Cureau et al. 2022). Figure 5 summarizes greenway design strategies to improve walking acoustic and thermal experience.

Reducing subjective noise loudness

Subjective noise perception and sound level increases negatively impact overall satisfaction, with subjective loudness being the primary factor. The key to enhancing greenway experiences is reducing noise's subjective loudness through direct attenuation and psychological strategies.

Direct noise reduction refers to the use of measures such as noise barriers to reduce sound levels. Acoustic barriers, using both air and solid mediums like vegetation (He 2020; Wang et al. 2022) and structures (Chen et al. 2022b), are effective for noise reduction. The material of landscape structures can be selected based on noise reduction and aesthetic requirements (He 2020). Vegetative barriers' efficacy depends on plant characteristics (Wang et al. 2022), with dense plantings notably lowering noise by 3.25 dB(A) at 20 m, whereas low vegetation offers a lesser 2 dB(A) reduction (Chen et al. 2022b). Plant configurations also improve psychological comfort (Ow and Ghosh 2017; He 2020). Therefore, dense and layered vegetation is advised for acoustic barriers, minimizing the solitary use of low-lying vegetation (Chen et al. 2003, 2022b; Ow and Ghosh 2017; Qing 2019; He 2020; Wang et al. 2022). It is recommended to combine fast-growing and slow-growing trees in acoustic barriers to achieve immediate and sustainable long-term effects (He 2020).

Additionally, psychological noise reduction can be achieved through the "masking effect" in acoustics. The "masking effect" suggests that the perception of one sound is diminished by the presence of another (Zhang 2016). Incorporating natural, positive sounds to overshadow negative noise can subjectively reduce noise loudness (Axelsson et al. 2014). As such, greenway design proposals should include bird-attracting and nectariferous plants, as well as dynamic water installations like fountains and cascades (Chen et al. 2022a, b; Guo et al. 2022; Qing 2019; Weng et al. 2021).

Differentiating design based on walking status

The regression analysis of noise and thermal parameters reveals that noise more significantly affects thermal

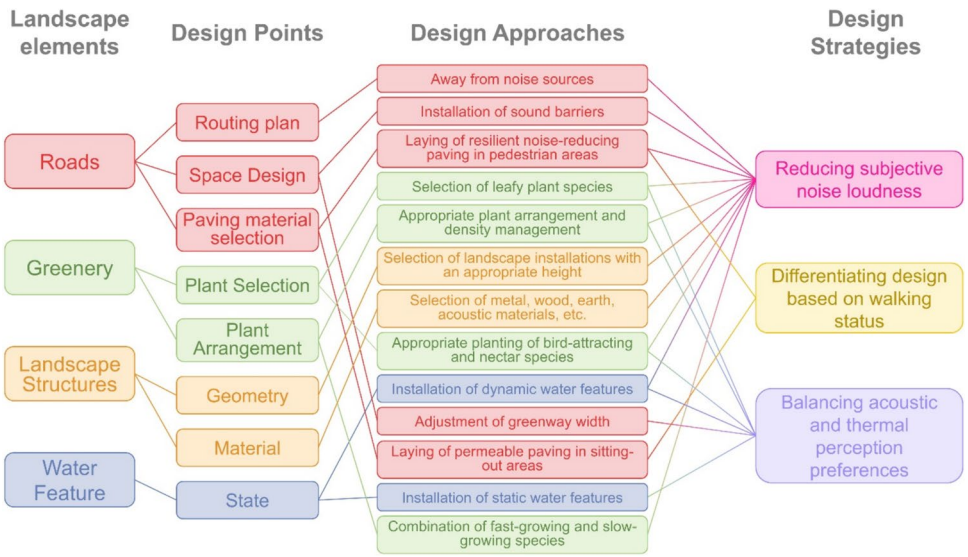


Fig. 5 Greenway design strategies to improve walking acoustic and thermal comfort

perception upon arrival, while after a 5-min break, thermal perception has a stronger influence on noise perception. Thus, greenway spaces should be designed to accommodate walking states. Walking paths should prioritize noise reduction to directly enhance overall satisfaction by mitigating noise perception during and immediately after walking, while rest areas should pay more attention to microclimate creation to indirectly improve satisfaction by enhancing thermal comfort and reducing noise perception after rest. Paving materials significantly affect air temperature at 0.4–0.6 m height, and permeable paving can reduce temperatures and increase humidity post-rainfall (Zeng and Dong 2014), making it ideal for greenway rest areas. Porous elastic road surfaces are recommended for greenway paths, offering both sports suitability and notable noise reduction (He 2020).

Balancing acoustic and thermal perception preferences

Enhancing greenway walking satisfaction requires the interplay of acoustic and thermal factors. This study proposes optimal noise level and air temperature ranges that elevate OES above the 0.5 threshold. Greenway design should thus balance the requirements of both acoustic and thermal aspects optimize user satisfaction.

Greenways with different initial noise levels should have distinct focuses during design or renovation. In areas with high noise levels or near municipal roads, priority should be given to noise reduction. For areas with lower noise levels or located within parks, it is also important to focus on the creation of a suitable microclimate, with a particular emphasis on managing air temperature.

Dense tree cover and confined greenways can offer shade and reduce air temperature, but may impede wind convection (Park et al. 2019; Liu et al. 2022), influencing sweat evaporation and thus thermal comfort (Liu et al. 2022). Water features help regulate local temperatures (Li et al. 2022) and enhance humidity (Nitidara et al. 2022), while flowing water can effectively mask noise (Axelsson et al. 2014). Integrating dynamic water elements along greenways can address both noise reduction and thermal comfort. Therefore, in greenway design, it is essential to carefully consider the impact of landscape elements on both acoustics and thermal comfort.

Limitations and future work

This study underscores the pivotal role of noise in greenway acoustics and its implications for subject comfort. The beneficial effects of positive sounds on comfort are a promising area for future investigation (Geng et al. 2022). Further research will extend our understanding of acoustic and thermal perceptions across different seasons during greenway walks.

Conclusion

In this study, greenway walking experiments were conducted, questionnaires and mobile measurements were combined to analyze the impact of the interplay between noise and thermal factors, and their impact on overall environmental satisfaction. The findings are compared with similar studies to investigate how acoustic-thermal elements affect perceived comfort in outdoor public spaces. Ultimately, this study proposed practical strategies for enhancing greenway-walking experience under the combined effects of noise and thermal environments. The main conclusions are as follows:

1) Noise and thermal factors interact, with their impact varying by walking state. Upon arrival, an increase in noise significantly deteriorates thermal comfort. Moreover, higher sound levels intensify warm thermal sensations, though this effect is not consciously perceived. While upon departure, the impact of noise on thermal perception is not statistically significant. In both walking states, thermal sensation significantly affects subjective noise perception, yet the trends of influence differ. Decreased thermal comfort increases the subjective noise loudness, but this effect is statistically significant only in the departure state. Microclimatic parameters have no impact on the subjective noise loudness.
2) During autumn greenway walks, acoustic factors exert a greater impact on overall environmental satisfaction, with subjective noise loudness being more influential than noise level, followed by air temperature. Greater noise decreases OES, while OES increases initially with T_a and then decreases. Moreover, the integrated effects of noise-thermal factors on OES show significant changes around 55–60 dBA. High OES is achieved when subjective noise loudness is relatively low, thermal sensation is warm, and the environment is nearly thermally comfortable.
3) To enhance the greenway-walking experience in autumn, the recommended parameters are ASL $\leq$ 59.12 dBA and 15.17 ℃ $\leq T_a \leq$ 18.75 ℃. Furthermore, three design strategies are proposed: reducing subjective noise loudness, differentiating design based on walking status, and balancing acoustic and thermal perceptual preferences.

Appendix

Appendix Questionnaire

1) Please describe your current thermal sensation:

□Very hot (4); □Hot (3); □Warm (2); □Slightly warm (1); □Neutral (0); □Slightly cool (-1); □Cool (-2); □Cold (-3); □Very cold (-4)

2) Please describe the comfort level of the current thermal environment:

□Comfortable (0); □Slightly uncomfortable (1); □Uncomfortable (2); □Very uncomfortable (3)

3) Please describe your subjective perception of the noise loudness:

□Hardly (1); □Relatively few (2); □Moderate (3); □More often (4); □Extremely much (5)

4) Please describe the visual aesthetics of the current environment:

□ Very Poorly (1); □ Poorly (2); □ Average (3); □ Good (4); □ Very good (5)

5) Do you enjoy the current landscape?

□ Not at all (-2); □ Somewhat don't enjoy it (-1); □ Neutral/fair (0) □ Somewhat enjoy it (1); □ Very enjoy it (2)

6) To what extent would you choose your current location as a destination for a greenway tour?

□ Would not choose it at all (-2); □Would not consider it very much (-1); □ Fair/neutral (0); □ Would consider it (1); □ Would choose it first (2)

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Data Availability Data will be made available on request.

Declarations

Conflict of interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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